

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.5: Respiration — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.5: Respiration
Driving Question	DRIVING QUESTION: How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Sealed Thermos of Uji That Keeps Changing — Predict & Anchor	A sealed thermos flask of fermented uji (uji wa kimea) brought from home to school in Kisumu produced visible bubbles and became increasingly sour over two hours, despite being completely sealed with no heat applied and no new ingredients added. When opened, a faint smell was noticeable. A side-by-side comparison with fresh uji wa kawaida (un-fermented maize porridge) in a sealed jar showed no such changes in the fresh sample.	Breathing (upumuaji) and cellular respiration (upumuaji wa seli) are two different processes. Breathing is the physical movement of air in and out of the lungs — visible and whole-body. Cellular respiration is a chemical process that occurs inside every living cell, releasing energy from food (substrate) and producing carbon dioxide and other products. The changes in the sealed uji are caused by cellular respiration carried out by living microorganisms already present in the fermented porridge — not by heat or external forces.	The bubbles and increasing sourness in the sealed thermos of fermented uji are produced by microorganisms (likely yeast and bacteria) carrying out cellular respiration inside the flask. These organisms break down sugars in the uji to release energy, producing carbon dioxide gas (the bubbles) and acids (the sour taste) as by-products. Because the flask is sealed and the process still occurs, we begin to ask: does this type of cellular respiration require oxygen, or can it happen without it? This question — aerobic versus anaerobic respiration — is what the rest of the unit will investigate and explain.
Lesson 2 Observe and Explore: Aerobic Respiration and the Energy in Our Food	Learners observed and read about the process of aerobic respiration, noting that glucose and oxygen are consumed inside the mitochondria of living cells to release energy (ATP), carbon dioxide, and water. They connected this to the energy needs of Amina the Nairobi sprinter and to the food journey from ugali to ATP in human cells.	Aerobic respiration requires glucose and oxygen as reactants and produces carbon dioxide, water, and ATP (energy). It occurs primarily in the mitochondria. A continuous supply of oxygen is essential for sustained energy release in active cells. The word equation is: Glucose + Oxygen yields Carbon Dioxide + Water + Energy (ATP).	Aerobic respiration begins to explain how living organisms release energy from food, including how a student's muscle cells use glucose from ugali during exercise. However, aerobic respiration alone does NOT fully explain the bubbling and souring inside the sealed uji thermos, because that flask has no fresh oxygen entering — something else must be happening, and that process will be investigated in Lesson

			3.
Lesson 3 Caught in the Act: Investigating Anaerobic Respiration in Yeast	Learners observed that Flask A (yeast + glucose) produced visible bubbles in the liquid and caused the lime water in the connected test tube to turn milky/white within 10–20 minutes. Flask B (glucose only, no yeast) produced no bubbles and the lime water remained clear. Learners also noted a faint smell (ethanol/yeasty) from Flask A but not from Flask B.	Learners learned that: (1) yeast carries out anaerobic respiration in the absence of oxygen; (2) the products are carbon dioxide (CO ₂) and ethanol; (3) CO ₂ is the gas that turns lime water milky — this is a standard chemical test for CO ₂ ; (4) the word equation is: glucose → ethanol + carbon dioxide (+ energy); (5) the yeast naturally present in fermented uji carries out this same process inside the sealed thermos, producing the bubbles Achieng observed.	Using today's experimental evidence, learners can now explain the anchoring phenomenon: the bubbles inside Achieng's sealed thermos of uji are carbon dioxide gas produced by yeast cells carrying out anaerobic respiration. The yeast consume glucose present in the fermented maize porridge, producing CO ₂ (the bubbles) and ethanol. No oxygen is required, which is why the process continues even inside a sealed thermos. The increasing sourness is not yet fully explained by today's lesson (lactic acid from bacterial fermentation will be addressed in a later lesson), giving learners a legitimate remaining question for the DQB.
Lesson 4 Anaerobic Respiration in Animals & Plants: Products, Equations, and the Sour Truth in Our Uji	Learners added universal indicator to three samples (fresh uji, fermented uji from the sealed thermos, and mala) and recorded colour changes and pH values. They observed the limewater turning milky in a sealed yeast-sugar flask with no oxygen input, confirming CO ₂ production under anaerobic conditions. Fermented uji registered pH 3–4 (orange-red with universal indicator), significantly lower than fresh uji (pH 5.5–6.5, yellow-green) and distilled water control (pH 7, green).	Anaerobic respiration in yeast and plants (alcoholic fermentation) converts glucose into ethanol and carbon dioxide, releasing a small amount of energy (ATP): $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. The CO ₂ produces the bubbles observed in the sealed thermos; the accumulation of acidic by-products lowers the pH of the uji, producing its sour taste. In animal muscle cells under oxygen-deficient conditions (e.g., a sprinting athlete), anaerobic respiration follows the lactic acid pathway: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$, producing lactic acid which causes the burning sensation in muscles. Both pathways produce far less ATP than aerobic respiration and do not require oxygen.	The sour taste and bubbles in the Kisumu student's sealed thermos of uji are explained by yeast cells carrying out alcoholic fermentation — anaerobic respiration — using glucose in the maize porridge as a substrate. Without oxygen, yeast cannot complete aerobic respiration, so it uses a shorter pathway that produces ethanol (responsible for any alcoholic notes) and CO ₂ (the bubbles). The accumulation of CO ₂ dissolved as carbonic acid, plus other organic acids, lowers the pH of the uji from approximately pH 6 to pH 3–4, which the tongue detects as sourness. The same principle — an acid product from anaerobic metabolism — explains why fresh uji (unfermented, no active yeast) does not sour in the same way, and why mala (fermented milk, produced by lactic acid bacteria) is similarly sour. In animal cells like those in a Nairobi sprinter's thigh muscles, a parallel but chemically distinct pathway produces lactic acid instead of ethanol, causing the characteristic burning sensation during intense exercise.
Lesson 5 Respiratory Substrates and Respiratory	We looked at data tables showing how much O ₂ is consumed and how much CO ₂ is produced when glucose, fat, and protein	The three main respiratory substrates are carbohydrates (RQ = 1.0), fats (RQ approximately 0.7), and proteins (RQ	The yeast in the sealed uji thermos is using glucose (a carbohydrate substrate) as its respiratory fuel. Because the flask is sealed

Quotient: What Fuel Are Our Cells Really Burning?	are completely oxidised during aerobic respiration. We also sorted familiar Kenyan foods (ugali, nyama choma, avocado, mala, mandazi dough, groundnut oil) into substrate categories.	approximately 0.8). The respiratory quotient (RQ) is calculated as the volume of CO ₂ produced divided by the volume of O ₂ consumed. An RQ above 1.0 indicates that anaerobic respiration is contributing to energy release, because fermentation produces CO ₂ without consuming O ₂ .	and oxygen becomes limited, the yeast shifts to anaerobic fermentation — this produces CO ₂ without consuming O ₂ , pushing the effective RQ above 1.0 and explaining why so many bubbles accumulate inside the sealed thermos even though no new ingredients were added.
Lesson 6 Factors Affecting the Rate of Respiration: Why Does Uji Ferment Faster in Mombasa Than in Kericho?	Yeast-glucose solutions at ~40 °C produced CO ₂ bubbles much faster (lime water turned milky in ~4 minutes) than solutions at ~5 °C (lime water barely changed in 15 minutes). Sealed (anaerobic) flasks showed slower initial CO ₂ output but increasing sourness over time. Higher glucose concentration (10 g/50 mL) produced faster and more sustained bubble rates than low concentration (2 g/50 mL). Germinating maize seeds turned lime water milky significantly faster than dry seeds of equal number, indicating higher respiration rates in metabolically active seeds.	The rate of respiration in living organisms — whether yeast in uji or cells in germinating seeds — is controlled by at least four key factors: (1) Temperature: higher temperatures increase enzyme activity and therefore increase respiration rate, up to an optimum beyond which enzymes denature; (2) Oxygen availability: aerobic respiration is faster and more efficient, but anaerobic respiration proceeds when oxygen is limited, producing acids and CO ₂ ; (3) Substrate concentration: more glucose (or other respiratory substrate) provides more fuel, increasing respiration rate up to enzyme saturation; (4) Surface area and metabolic activity: germinating seeds have greater metabolic activity and surface area for gas exchange, producing more CO ₂ per unit time than dormant dry seeds. These factors explain why identical uji ferments noticeably faster on a warm Mombasa afternoon (~30–35 °C) than during a cool Kericho highland morning (~13–16 °C) — the higher temperature accelerates yeast enzyme activity, increasing the rate of anaerobic fermentation, CO ₂ production, and acid accumulation.	The rate of anaerobic respiration (fermentation) in the yeast cells inside the sealed uji thermos is directly regulated by the temperature of the surroundings, the concentration of glucose in the uji, and the availability of oxygen diffusing through the porridge. At the warmer coastal temperatures of Mombasa, yeast enzymes catalyse the breakdown of glucose to ethanol and carbon dioxide at a faster rate — producing more bubbles and more lactic/acetic acid (sourness) per unit time. In the cooler Kericho highlands, the same enzymes work more slowly, reducing the fermentation rate and preserving the uji's freshness longer. This also explains why sealed, glucose-rich uji ferments faster than fresh unsweetened uji wa kawaida — freshly cooked uji has few or no live fermenting microorganisms (cooking kills them), and where microorganisms are present a higher substrate concentration directly fuels enzyme activity. Understanding and controlling these factors has direct economic value: food processors in Nairobi can maintain optimal fermentation temperatures to standardise uji, yoghurt, and togwa production; biogas plants in the Rift Valley can maximise methane output by controlling substrate concentration and temperature; and farmers can store grain safely by keeping it cool and dry to suppress microbial respiration.
Lesson 7 DQB Revision & Economic Importance of Anaerobic Respiration	Learners colour-coded their Driving Question Board, conducted a gallery walk with evidence citations, and listened to peer group presentations on five Kenyan economic applications of anaerobic respiration (mala/yoghurt, uji fermentation,	Anaerobic respiration is not only a cellular survival mechanism — it is the biochemical engine behind major sectors of Kenya's agro-processing economy, from Brookside dairy to Rift Valley biogas plants to mandazi sellers in Mombasa. The same CO ₂ that	Learners can now explain how anaerobic respiration products — CO ₂ , lactic acid, and methane — are harnessed in specific Kenyan industries, name the microorganisms involved, write the relevant word equations, and connect each

	Rift Valley biogas, silage, mandazi leavening).	bubbled in the sealed uji thermos causes mandazi dough to rise; the same lactic acid that soured the uji is what preserves mala and silage. Each application involves specific microorganisms carrying out anaerobic respiration under oxygen-limited conditions, producing economically valuable by-products. Brewing alcoholic substances is explicitly excluded from responsible application of this knowledge.	application to the anchoring phenomenon of the fermenting uji thermos and to Kenya's Vision 2030 agro-processing goals.
Lesson 8 Model Building & Project Showcase: Final Explanation of Respiration	Learners observed three value-addition projects — yoghurt, uji/porridge fermentation, and a biogas model — displayed as gallery walk stations. They observed physical evidence of anaerobic respiration: pH change (colour shift of red cabbage indicator in fermented uji), gas collection in a balloon (biogas model), and texture/smell change in yoghurt. They observed two Lesson 1 naive models displayed alongside two Final Models on the board, making the visual contrast between initial and revised scientific thinking concrete.	Learners consolidated that: (1) both aerobic and anaerobic respiration release ATP from organic substrates; (2) the respiratory quotient ($RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}}$) identifies substrate type — $RQ = 1$ for carbohydrates, < 1 for fats, > 1 for anaerobic fermentation; (3) factors such as temperature, substrate concentration, enzyme availability, and oxygen levels control the rate of both pathways; (4) anaerobic fermentation by yeast and lactic acid bacteria produces CO_2 , ethanol, and lactic acid — precisely the agents responsible for the bubbles and sourness in the sealed uji thermos; (5) these same processes have direct economic value in yoghurt production, fermented porridge (uji wa kimea), and biogas generation in Kenyan households and industries.	Learners constructed a Final Annotated Explanatory Model linking the anchoring phenomenon (sealed thermos of uji, Kisumu student, bubbles, sour smell) to all unit concepts: aerobic and anaerobic respiration pathways, respiratory substrates and RQ values, factors affecting respiration, and economic applications. They closed every outstanding question on the Driving Question Board with evidence-based answers and articulated a full 3-sentence response to the driving question: 'How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?'

ATTRIBUTION AND LICENSING

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